



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)

UNIT TEST

21-02-2021

JEE(Main + Advanced) : NURTURE PHASE-I(B) & NURTURE LIVE PHASE-I(B)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	D	C	A	A,C,D	A,B,C	B,D	A,D	A,D
SECTION-III	Q.	11	12	13							
	A.	C	B,C,D	A,C,D							
SECTION-III	Q.	1	2	3	4	5					
	A.	4	0	5	2	4					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	A	B	C	A,B,C,D	A,B	A,B,C,D	A,C,D	A,B,C
SECTION-III	Q.	11	12	13							
	A.	A,B,D	B,C,D	A,B							
SECTION-III	Q.	1	2	3	4	5					
	A.	4	2	4	1	1					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	A	A	A,B,D	A,C,D	B,D	B,C,D	A,C
SECTION-III	Q.	11	12	13							
	A.	B,C,D	A,C,D	A,B,C							
SECTION-III	Q.	1	2	3	4	5					
	A.	7	5	5	0	7					

JEE(Main + Advanced) : NURTURE PHASE-I(B) & NURTURE LIVE PHASE-I(B)**PART-1 : PHYSICS****SOLUTION****SECTION-I****1. Ans. (A)****Sol.** Let $\rho = kR^x g^y G^z$

$$[M^1 L^{-3} T^0] = K[L]^x [LT^{-2}]^y [M^{-1} L^3 T^{-2}]^z$$

$$\Rightarrow [M^1 L^{-3} T^0] = K[L]^{x+y+3z} [M]^{-z} [T]^{-2y-2z}$$

Comparing exponents on both sides

$$-z = 1 \Rightarrow z = -1$$

$$\text{and } -2y - 2z = 0 \Rightarrow 2y = -2z \Rightarrow y = +1$$

$$\text{and } x + y + 3z = -3 \Rightarrow x + 1 - 3 = -3$$

$$\Rightarrow x = -1$$

$$\therefore \rho = kR^{-1} g G^{-1} \Rightarrow \rho = \frac{\text{kg}}{\text{RG}}$$

2. Ans. (C)**Sol.** As we know that

$$|A - B| \leq |\vec{A} + \vec{B}| \leq A + B$$

for given question

$$|12 - 5| \leq |\vec{A} + \vec{B}| \leq 12 + 5$$

$$7 \leq |\vec{A} + \vec{B}| \leq 17$$

Hence Ans. (C)

3. Ans. (D)**Sol.** Power = $[ML^2 T^{-3}]$

$$n_1 u_1 = n_2 u_2$$

$$n_1 = \left[\frac{M_1}{M_2} \right] \left[\frac{L_1}{L_2} \right]^2 \left[\frac{T_1}{T_2} \right]^{-3}$$

$$n_1 = \frac{\gamma^3}{\alpha \beta^2}$$

4. Ans. (C)

$$\text{Sol. } x = \frac{v_0}{\alpha} (1 - e^{-\alpha t})$$

$$[\alpha t] = 1 \Rightarrow [\alpha] = \frac{1}{[T]} \Rightarrow [\alpha] = [T^{-1}]$$

$$\text{and } [x] = \left[\frac{v_0}{\alpha} \right] \Rightarrow [v_0] = [\alpha][x]$$

$$\Rightarrow [v_0] = [LT^{-1}]$$

$$\therefore [v_0] = [M^0 L^1 T^{-1}] \text{ and } [\alpha] = [M^0 L^0 T^{-1}]$$

5. Ans. (A)**Sol.** $[Y] = [V]^x [A]^y [F]^z$

$$\Rightarrow [ML^{-1} T^{-2}] = [LT^{-1}]^x [LT^{-2}]^y [MLT^{-2}]^z$$

$$\Rightarrow [ML^{-1} T^{-2}] = [L]^{x+y+z} [T]^{-x-2y-2z} [M]^z$$

Comparing exponents : $z = 1$

$$x + y + z = -1 \Rightarrow x + y = -2 \dots (i)$$

$$\text{and } -x - 2y - 2z \Rightarrow -x - 2y - 2 = -2 \dots (ii)$$

$$\Rightarrow x + 2y = 0 \dots (ii)$$

Solving (i) and (ii)

$$x = -4 \text{ and } y = 2$$

$$\therefore [Y] = [FA^2 V^{-4}]$$

6. Ans. (A,C,D)**Sol.** Physical Quantities with different dimensions cannot be added or subtracted.**7. Ans. (A,B,C)****Sol.** $E = Ae^{-Bx}$

$$[E] = [A] = ML^2 T^{-2} = [\text{Power} \times \text{time}]$$

$$[Bx] = 1$$

$$[B] = \frac{1}{[x]} = \frac{1}{L} = L^{-1} = \frac{1}{\text{velocity} \times \text{time}}$$

ALLEN

8. Ans. (B,D)

Sol. m, ℓ, v

$$v \propto m^x \ell^y$$

$$[LT^{-1}] \propto [M^x L^y]$$

→ No solution for x, y which means all three physical quantities are independent

(B) P, ρ, v

$$P \propto \rho^x v^y$$

$$[ML^{-1}T^{-2}] = [ML^{-3}]^x [LT^{-1}]^y$$

$$x = 1$$

$$-3x + y = -1 \Rightarrow y = 2$$

$$-2 = -y \Rightarrow y = 2$$

→ All three quantities are dependent hence, they cannot be a set of fundamental quantities.

(C) F, v, t

$$F \propto v^x t^y$$

$$[MLT^{-2}] = [LT^{-1}]^x [T]^y$$

→ No solution for x, y which means all three physical quantities are independent

(D) F, p, t

$$F = \frac{p}{t}$$

→ All three quantities are dependent hence, they cannot be a set of fundamental quantities.

9. Ans. (A,D)

Sol. Refer Class Notes.

10. Ans. (A,D)

Sol. $y = a \sin(bt - x)$

∴ $\sin\theta$ is dimensionless

$$\therefore [y] = [a]$$

' θ ' is also dimensionless and we can subtract same dimension quantity.

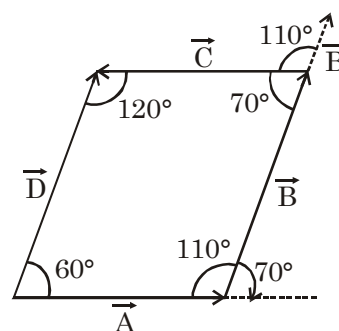
$$\therefore [bt] = [cx] = 1$$

$$\Rightarrow [b] = [T^{-1}] \text{ and } [C] = [L^{-1}]$$

$$\text{dimension of } \left[\frac{b}{c} \right] = \left[\frac{T^{-1}}{L^{-1}} \right] = [LT^{-1}] = [v]$$

11. Ans. (C)

Sol. As we know that angle between two vectors is minimum angle between heads or tails of two vector from given diagram



Angle between $\vec{A}$ and $\vec{B}$ is 70°

Angle between $\vec{B}$ and $\vec{C}$ is 110° .

Angle between $\vec{C}$ and $\vec{D}$ is 120° .

12. Ans. (B, C, D)

$$\text{Sol. } \vec{A} + \vec{B} = -\vec{C}$$

$$\vec{A} + \vec{B} + \vec{C} = 0$$

$$\vec{C} + \vec{A} = -\vec{B}$$

$$|\vec{C} + \vec{A}| = |\vec{B}|$$

13. Ans. (A, C, D)

$$\text{Sol. Given } I = a \left(1 - e^{-\frac{t}{RC}} \right)$$

Input and output of exponential function i.e. e^x and x both are dimensionless.

$$\therefore [I] = [a]$$

$$\text{and } \left[\frac{t}{RC} \right] = 1 \Rightarrow [RC] = [t]$$

Given $P = I^2 R$

$$\therefore R = \frac{P}{I^2}$$

We know that $P = \frac{\text{Energy}}{t}$

$$[P] = \frac{[ML^2T^{-2}]}{[T]} = [ML^2T^{-3}]$$

$$\therefore [R] = \frac{[P]}{[I]^2} = \frac{[ML^2T^{-3}]}{[I]^2} = [ML^2T^{-3}I^{-2}]$$

SECTION-III

1. **Ans. 4**

$$\text{Sol. } |\hat{A} + \hat{B}| = \sqrt{1^2 + 1^2 + 2(1)(1)\cos 60^\circ} = \sqrt{3} = \sqrt{x}$$

$$\therefore x = 3$$

$$\text{and } |\hat{A} - \hat{B}| = \sqrt{1^2 + 1^2 - 2\cos 60^\circ} = 1 = \sqrt{y}$$

$$\therefore y = 1$$

$$\text{So, } x + y = 4$$

2. **Ans. 0**

$$\text{Sol. } \frac{yt}{v} = \text{dimensionless; } \frac{yt}{v} = 1;$$

$$y = \frac{v}{t} \Rightarrow \frac{L^1T^{-1}}{T^1} \Rightarrow MT^{-2} \Rightarrow M^0L^1T^{-2}$$

$$\text{Value of } x + 2y + z \Rightarrow 0 + 2 - 2 \Rightarrow 0$$

3. **Ans. 5**

$$\text{Sol. } F = \frac{\alpha}{\beta + \rho}$$

$$MLT^{-2} = \frac{\alpha}{ML^{-3}}$$

$$\Rightarrow \alpha = M^2L^{-2}T^{-2}$$

$$\beta = ML^{-3}$$

$$\therefore (-2 - 3) = 5$$

4. **Ans. 2**

$$\text{Sol. } [E] = ML^2T^{-2}$$

$$n = 5 \left(\frac{M_1}{M_2} \right) \left(\frac{L_1}{L_2} \right)^2 \left(\frac{T_1}{T_2} \right)^{-2} = 5(2) \left(\frac{1}{2} \right)^2 \left(\frac{1}{2} \right)^{-2}$$

5. **Ans. 4**

$$\text{Sol. } P \propto h^\alpha C^\beta G^\gamma$$

$$P = kn^\alpha C^\beta G^\gamma$$

$$MC^2T^{-3} = [ML^2T^{-1}]^\alpha [LT^{-1}]^\beta [M^{-1}L^3T^{-2}]^\gamma$$

on comparison

$$1 = \alpha - \gamma \quad \dots\dots(1)$$

$$2 = 2\alpha + \beta + 3\gamma \quad \dots\dots(2)$$

$$-3 = -\alpha - \beta - 2\gamma \quad \dots\dots(3)$$

on solving equation (1), (2) & (3)

$$\alpha = 0$$

$$\beta = 5$$

$$\gamma = -1$$

$$\alpha + \beta + \gamma = 4$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(B)
2. Ans.(B)
3. Ans.(A)
4. Ans.(B)
5. Ans.(C)
6. Ans.(A,B,C,D)
7. Ans.(A, B)
8. Ans.(A, B, C, D)
9. Ans.(A, C, D)
10. Ans.(A,B,C)
11. Ans.(A,B,D)
12. Ans.(B,C, D)
13. Ans.(A,B)

SECTION-III

1. Ans.(4)
2. Ans.(2)
3. Ans.(4)
4. Ans.(1)
5. Ans.(1)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A)
Sol. $(x-2)^2 + |x-2| = 0$
 $(x-2)^2 = 0$ and $|x-2| = 0$
 $x = 2$ and $x = 2$
 $\therefore x = 2$
 No. of solutions = 1

2. Ans. (C)

Sol.
$$\frac{(\sqrt{x}-\sqrt{y})^2 + 4\sqrt{xy}}{x-y} = \frac{7}{3}$$

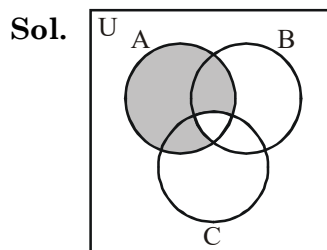
$$\Rightarrow \frac{(\sqrt{x}+\sqrt{y})^2}{(\sqrt{x}-\sqrt{y})(\sqrt{x}+\sqrt{y})} = \frac{7}{3}$$

$$\Rightarrow \frac{\sqrt{x}+\sqrt{y}}{\sqrt{x}-\sqrt{y}} = \frac{7}{3}$$

$$\Rightarrow \frac{2\sqrt{x}}{2\sqrt{y}} = \frac{10}{4}$$

$$\frac{x}{y} = \frac{25}{4}$$

3. Ans. (A)



4. Ans. (A)

Sol. $n(M \cap P) = 40$
 $n(P \cap C) = 30$
 $n(C \cap M) = 50$

$$n(\overline{M \cup P \cup C}) = 20$$

$$\text{Let } n(M \cap P \cap C) = x$$

$$n(M \cup P \cup C) = U - n(\overline{M \cup P \cup C})$$

$$= 200 - 20 = 180$$

$$n(M \cup P \cup C) = n(M) + n(P) + n(C) - n(M \cap P) - n(P \cap C) - n(C \cap M) + n(M \cap P \cap C)$$

$$\Rightarrow 180 = 120 + 90 + 70 - 40 - 30 - 50 + x$$

$$\Rightarrow x = 20$$

5. **Ans. (A)**

Since $x^2 - y^2$ is even $\Rightarrow$ either both x & y are even or both x & y are odd

$\Rightarrow x + y$ and $x - y$ both are even in both the cases

$\Rightarrow x^2 - y^2$ is divisible by 4

but 2010 is not divisible by 4 hence no solution.

6. **Ans. (A,B,D)**

$$\text{Sol. } \sqrt{2} - \sqrt{1} + \sqrt{3} - \sqrt{2} + \sqrt{4} - \sqrt{3} \dots + \sqrt{9} - \sqrt{8}$$

$$= \sqrt{9} - \sqrt{1} = 3 - 1 = 2$$

7. **Ans. (A,C,D)**

Sol. We have, $A' \cap B' = (A \cup B)'$

$$\therefore n(A' \cap B') = n((A \cup B)') = n(U) - n(A \cup B)$$

$$= n(U) - [n(A) + n(B) - n(A \cap B)]$$

$$= 700 - (200 + 300 - 100) = 300.$$

8. **Ans. (B,D)**

Sol. (A) $\{5\} \in A$ is wrong since there is no element $\{5\}$ in set A

(B) $\{5, 7\}$ is element of A . So $\{5, 7\} \in A$

(C) $\{1, 2, 3\}$ is subset of A

(D) $\{\{5, 7\}\}$ is subset of A

9. **Ans. (B,C,D)**

$$\text{Sol. } \frac{1}{A+B} + \frac{1}{B+C} + \frac{1}{C+A} = \frac{7}{10}$$

Multiply both sides by $A + B + C$

$$\frac{(A+B)+C}{A+B} + \frac{A+(B+C)}{B+C} + \frac{B+(C+A)}{C+A}$$

$$= \frac{7}{10}(A+B+C)$$

$$\Rightarrow \left(1 + \frac{C}{A+B}\right) + \left(\frac{A}{B+C} + 1\right) + \left(\frac{B}{C+A} + 1\right)$$

$$= \frac{7}{10}(7)$$

$$\Rightarrow \frac{A}{B+C} + \frac{B}{C+A} + \frac{C}{A+B} = \frac{49}{10} - 3 = \frac{19}{10}$$

10. **Ans. (A,C)**

$$\text{Sol. } \sqrt{11+3+2+2\sqrt{11}(-\sqrt{3})+2(-\sqrt{3})\sqrt{2}+2\sqrt{11}\sqrt{2}}$$

$$= \sqrt{(\sqrt{11} - \sqrt{3} + \sqrt{2})^2}$$

$$= \sqrt{11} - \sqrt{3} + \sqrt{2}$$

11. **Ans. (B,C,D)**

$$\text{Sol. } a + b + c = 3, a^2 + b^2 + c^2 = 13, a^3 + b^3 + c^3 = 27$$

$$(a + b + c)^2 = (a^2 + b^2 + c^2) + 2(ab + bc + ca)$$

$$\Rightarrow 9 = 13 + 2(ab + bc + ca)$$

$$\Rightarrow ab + bc + ca = -2 \quad \dots(1)$$

$$a^3 + b^3 + c^3 - 3abc$$

$$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$\Rightarrow 27 - 3abc = (3)(13 - (-2)) \quad \text{from (1)}$$

$$\Rightarrow abc = -6$$

$$\text{Now } \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{bc + ca + ab}{abc} = \frac{-2}{-6} = \frac{1}{3}$$

12. Ans. (A,C,D)

Sol. Case-I : $x^2 - 5x + 5 = 1 \Rightarrow x = 1, 4$

Case-II : $x^2 + 4x - 60 = 0 \Rightarrow x = -10, 6$

Case-III : $x^2 - 5x + 5 = -1 \Rightarrow x = 2, 3$

Now at $x = 2$, $x^2 + 4x - 60$

$$= 4 + 4(2) - 60 = -48 \text{ i.e. even}$$

$\therefore x = 2$ is included

Now at $x = 3$, $x^2 + 4x - 60$

$$= 9 + 12 - 60 = -39 \text{ i.e. odd}$$

$\therefore x = 3$ is rejected

$$\text{Sum} = 1 + 4 + 2 + (-10) + 6 = 3$$

13. Ans. (A,B,C)

Sol. Let $f(x) = ax^3 + bx^2 + cx + d$

$$f(1) = 0 \text{ and } f(-2) = 0$$

$\therefore x - 1$ and $x + 2$ are factors

$$\therefore f(x) = (x - 1)(x + 2)(px + q)$$

$$\text{Now } f(-1) = (-1 - 1)(-1 + 2)(-p + q)$$

$$4 = -2(-p + q) \Rightarrow p - q = 2 \dots (1)$$

$$\text{Now } f(2) = (2 - 1)(2 + 2)(2p + q)$$

$$28 = 4(2p + q) \Rightarrow 2p + q = 7 \dots (2)$$

Add eq. (1) & (2)

$$3p = 9 \Rightarrow p = 3 \text{ and } q = 1$$

$$\therefore f(x) = ax^3 + bx^2 + cx + d$$

$$= (x - 1)(x + 2)(3x + 1)$$

$$f(0) = d = (0 - 1)(0 + 2)(0 + 1) = -2$$

SECTION-III

1. Ans. 7

$$\text{Sol. } \frac{3}{x} + \frac{5}{y} = \frac{21}{xy} + 1$$

$$\Rightarrow 3y + 5x = 21 + xy ; (x, y \neq 0)$$

$$\Rightarrow xy - 5x - 3y + 21 = 0$$

$$\Rightarrow x(y - 5) - 3(y - 5) + 6 = 0$$

$$\Rightarrow (x - 3)(y - 5) = -6$$

$$\therefore (x - 3, y - 5) \equiv (1, -6), (-1, 6)$$

$$(2, -3), (-2, 3)$$

$$(3, -2), (-3, 2)$$

$$(6, -1), (-6, 1)$$

$$\therefore (x, y) \equiv (4, -1), (2, 11)$$

$$(5, 2), (1, 8)$$

$$(6, 3), (0, 7) \rightarrow (\text{reject}) (\because x \neq 0)$$

$$(9, 4), (-3, 6)$$

number of possible ordered pairs (x,y) is 7

2. Ans. 5

Sol. Number is divisible by 9 and 8

The Number is divisible by 8 if last three digits are divisible by 8 i.e. 79b is divisible by 8 $\Rightarrow b = 2$

For divisible by 9

$$a + 6 + 7 + 9 + b \text{ is divisible by } 9$$

$$\Rightarrow a + 24 \text{ is divisible by } 9$$

$$\Rightarrow a = 3$$

$$\Rightarrow a + b = 5$$

3. Ans. 5

$$\text{Let } P(x) = -\frac{1}{4}(x-1)(x-2)(x-3) + ax^2 + bx + c$$

$$\text{Given } P(1) = 1, P(2) = 2, P(3) = 3$$

$$\Rightarrow a + b + c = 1, 4a + 2b + c = 2 \text{ \& }$$

$$9a + 3b + c = 3$$

Solving these we get $a = c = 0$ & $b = 1$

$$\therefore P(x) = -\frac{1}{4}(x-1)(x-2)(x-3) + x$$

$\Rightarrow$ Remainder when $P(x)$ is divided by $x + 1$ is

$$P(-1) = -\frac{1}{4}(-2)(-3)(-4) - 1 = 6 - 1 = 5$$

4. Ans. 0

Sol. Let $x = a - \frac{1}{a} \Rightarrow x^3 = \left(a - \frac{1}{a}\right)^3$

$$\Rightarrow x^3 = a^3 - \frac{1}{a^3} - 3.a.\frac{1}{a}\left(a - \frac{1}{a}\right)$$

$$\Rightarrow x^3 + 3x = a^3 - \frac{1}{a^3}$$

$$= (7 + 5\sqrt{2}) - \frac{1}{7 + 5\sqrt{2}}$$

$$= (7 + 5\sqrt{2}) - \left(\frac{7 - 5\sqrt{2}}{49 - 50}\right)$$

$$= 7 + 5\sqrt{2} + 7 - 5\sqrt{2}$$

$$\Rightarrow x^3 + 3x - 14 = 0$$

5. Ans. 7

Sol. $= \sqrt{(\sqrt{6})^2(\sqrt{5})^2 - 2\sqrt{6}\sqrt{5}} - \sqrt{(\sqrt{6})^2 + (2)^2 - 2(\sqrt{6})(2)}$

$$= \sqrt{(\sqrt{6} - \sqrt{5})^2} - \sqrt{(\sqrt{6} - 2)^2}$$

$$= |\sqrt{6} - \sqrt{5}| - |\sqrt{6} - 2|$$

$$= \sqrt{6} - \sqrt{5} - (\sqrt{6} - 2)$$

$$= 2 - \sqrt{5} = \alpha - \sqrt{\beta}$$

$$\therefore \alpha = 2 \text{ and } \beta = 5$$